

An adaptive role for anhedonia following early-life unpredictability

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Abstract

Early-life unpredictability (ELU), a form of early-life adversity, has been shown to lead to asymmetrically greater impact of negative, rather than positive, outcomes during learning. Separately, work on consummatory anhedonia suggests that reduced pleasure can arise from “warping” internal representations of environmental states, such that outcomes are perceived as more distant and prediction errors are reduced. Although ELU and anhedonia are highly correlated in the population, how these mechanisms interact remains unclear. Here, we use simulations to examine whether adopting an anhedonic belief about the state space can attenuate the hedonic asymmetry associated with ELU. Combining these two computational frameworks, we show that warping reduces perceived hedonic experience while reshaping the balance between gains and losses. While belief-space warping attenuates both positive and negative hedonic experience, it disproportionately suppresses negative outcomes and alters the tradeoff structure. These results suggest that anhedonic belief representations may mitigate the subjective impact of negative outcomes under ELU under specific environmental structures, offering a computational account of how these processes jointly shape hedonic experience. More broadly, the work provides a minimal normative framework for understanding when belief-space warping may confer local advantages, while also highlighting important limitations for translation to real-world cognition.

Introduction

Anhedonia, a reduction in experienced pleasure, is a transdiagnostic symptom found in many mental health conditions, ranging from depression to schizophrenia (Pizzagalli, 2022). Recently, a computational model was proposed that explained this *consummatory* anhedonia as resulting from individuals adopting a “warped” view of environmental contingencies, such that rewarding outcomes seemed more distant and thus muting the hedonic experience (RPEs; Reward Prediction Errors) resulting from their pursuit. Importantly, this account provides a computational-level explanation of anhedonia, rather

than a complete psychological or biological model of the phenomenon.

This framework captures a wealth of observations in anhedonic individuals across diagnoses, and provides meaningful suggestions for treatments that focus on correcting the distorted belief structure. It also raises the question of *why* individuals would adopt such a mindset, given that it reduces subjective experience of pleasure and also overall outcomes.

Here, we examine how, for at least one population with high prevalence of anhedonia, adopting such a mindset may alter the balance of hedonic experience in a locally desirable manner, particularly in certain environments. Specifically, we examine the case of individuals with high early-life unpredictability (ELU; (Baram et al., 2012; Glynn & Baram, 2019; Glynn et al., 2019)). ELU, a form of early-life adversity, has been shown to predict adverse mental-health outcomes, differences in reward processing and decision-making, and memory deficits, in each case explaining additional variance over and above other commonly observed forms of adversity such as socioeconomic status or childhood and life-long trauma (Chen et al., 2024; Doom et al., 2016, 2024; Glynn & Baram, 2019). Of particular relevance is the fact that ELU has been shown to correlate strongly with the prevalence of anhedonia (Hunt et al., 2024; Spadoni et al., 2022). A recent computational model offered an efficient coding explanation for many behavioral phenotypes shown to result from ELU (Harhen & Bornstein, 2024). The model showed that efficient coding of ELU should result, in adulthood, in persistent enhanced learning from punishments along with diminished learning from rewards. In the present work, we use this asymmetry as a minimal computational proxy for ELU-related biased updating under uncertainty. In this framework, individuals with high ELU adapt, during a sensitive period of high plasticity, to the associative variability in their environment by adopting tuning curves that “smear” expectations across time. The resulting extended uncertainty produces an outsized prediction error response to omitted or punishing value signals, and a muted re-



sponse to positive associations or rewards. Here, we examine in a simulation study how the hedonic experience that results from this asymmetric outcome weighting is altered by two distinct methods for belief-space warping previously proposed to explain anhedonia. We find that, for simulated high-ELU agents, the increase in perceived distance to goal *benefits* them in that it also asymmetrically reduces negative reward prediction errors, which have been linked to hedonic experience (Eldar et al., 2016). This combination suggests that belief-space warping may confer conditional advantages for high-ELU individuals in environments where negative and positive outcomes are in close competition. We close by discussing potential implications for treatment.

Method

Simulation and Setup

All simulations and analyses were performed in MATLAB R2024a, and consisted of agents traversing a “grid-world” environment with sparse rewards, detailed below. To model Early Life Unpredictability (ELU), we implemented a learning-rate asymmetry in a temporal-difference reinforcement learning agent (Harhen & Bornstein, 2024), where learning-rate asymmetry is treated as a minimal computational proxy for biased updating under uncertainty. Specifically, the negative learning rate was constrained to be strictly greater than or equal to the positive learning rate, consistent with the previous finding that ELU agents should normatively learn to disproportionately update from negative outcomes. Positive and negative learning rates were linked by a single asymmetry parameter A , which controlled the relative weighting between the two rates. For a given net learning rate S , the positive and negative learning rates were computed as:

$$\alpha^+ = \frac{S}{1+A}, \quad \alpha^- = \frac{A*S}{1+A}$$

such that $\alpha^- \geq \alpha^+$ when $A \geq 1$, and $\alpha^- + \alpha^+ = S$.

The asymmetry parameter A varied across values $\{1, 1.5, 3, 10, 20, 50, 100\}$, and the total learning rate S was set to $\{0.1, 0.3, 0.5, 0.7\}$. Varying S allowed us to examine whether the effects of learning asymmetry depended on the overall magnitude of learning.

Agents followed a standard temporal-difference reinforcement learning algorithm. State values were updated according to:

$$V(s_t) \leftarrow V(s_t) + \alpha^\pm \delta_t$$

where the Reward Prediction Error (RPE; δ_t) was defined as

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t).$$

Following the previous framework (Hall et al., 2024), instantaneous reward r_t was defined as progress toward

the goal, defined as the reduction in distance-to-goal between successive states,

$$r_t = d(s_t) - d(s_{t+1}),$$

where d is the manhattan distance from current state to goal state and such that positive reward corresponds to movement closer to the goal and negative reward reflects movement away from the goal. State values were initialized to zero at the beginning of each simulation. Action selection followed a standard softmax policy with inverse temperature parameter fixed at 1 across all simulations. The action space consisted of $[left, right]$ in the one-dimensional environment, and $[left, right, up, down]$ in the two-dimensional environment.

Also as in the simulation environment set up by (Hall et al., 2024), we simulated three task conditions: one control condition (unaltered, ground-truth state-space) and two *belief-warping* conditions (*dimensionality addition* and *rejection*). Each was simulated at each of the varying levels of ELU-induced reward outcome asymmetry A and overall learning rate S . Specifically, for each condition, at each level of $[A, S]$, 3000 agents were simulated with temporal discount factors γ randomly sampled from a uniform distribution $[0.50, 0.99]$.

In the control condition (Fig. 1A, left), the agents navigated a one-dimensional grid-world consisting of 10 states. In the dimensionality-addition condition (Fig. 1A, middle), an additional spatial dimension of the same length was randomly introduced during learning, expanding the belief space to a two-dimensional grid-world and increasing the inferred distance to the goal without altering reward availability. The manipulation can be interpreted as increasing representational complexity by introducing additional latent structure. In the rejection-backward condition (Fig. 1A, right), agents attempting to move into the goal state were probabilistically teleported backward to a previously visited state with probability sampled from $\{0.2, 0.5, 0.8\}$, introducing belief-level setbacks that increase the effective distance to the goal. This manipulation can be interpreted as introducing stochastic setbacks or uncontrollability within the agent’s belief structure.

Each simulated episode ended when it reached the goal state, with maximum steps capped at 120. Prediction errors were accumulated only over non-terminal transitions prior to goal attainment.

Data Analysis

We defined hedonic experience as the sum of Reward Prediction Error (δ) arising during temporal-difference learning across an episode, consistent with prior computational accounts of consummatory experience (Hall et al., 2024). Positive and negative hedonic experience were analyzed separately to capture valence-level asymmetries in experience.

To compute subject-level summary statistics for hedonic experience (h), prediction errors were normalized by the total reward magnitude obtained by each agent. This normalization ensured comparability across conditions, consistent with the assumption in the previous framework (Hall et al., 2024) that total reward is held constant across experimental manipulations.

$$h = \frac{\mathbb{E}[\delta]}{\mathbb{E}[|Reward|]}$$

We quantified subject-level hedonic asymmetry using the following metric:

$$\text{Asymmetry Index} = \frac{\mathbb{E}[h^+] - |\mathbb{E}[h^-]|}{\mathbb{E}[h^+] + |\mathbb{E}[h^-]|}$$

where $\mathbb{E}[h^+]$ denotes the expected positive hedonic experience, and $\mathbb{E}[h^-]$ denotes the expected negative hedonic experience.

Results

Belief-space warping increases perceived distance to goal

We first verified that the manipulations of belief space actually increased the effective distance to the goal, quantified as the number of steps required to reach the terminal state (Fig. 1B). Pooling across all values of S and A , the three conditions exhibited markedly different overall performance. Control trajectories converged rapidly (mean ≈ 12 steps; median = 11), Dim-add substantially increased path length (mean ≈ 45 steps; median = 43), and Reject produced the longest trajectories (mean ≈ 108 steps), with a large fraction of episodes terminating at the maximum allowed steps. These distributional shifts are consistent with the original model’s characterization of anhedonia, with Dim-add expanding the latent belief space and Reject introducing belief-level setbacks that systematically impede progress toward the goal.

Belief-space warping reduces mean hedonic outcomes

We next quantified hedonic experience using the sum of reward prediction error δ , normalized by total reward magnitude to ensure comparability across conditions (Fig. 2 A, C, E). Pooling across S and A and estimating uncertainty via subject-level bootstrapping (1000 resamples; 95% CI), belief-space warping produced large reductions in mean per-step hedonic experience (h): compared with Control (mean = 0.7883, CI [0.7872, 0.7895]), Dim-add reduced h by $\Delta = -0.1461$ (CI [-0.1475, -0.1448]; -18.53% , $d = -1.02$), whereas Reject reduced h by $\Delta = -0.7194$ (CI [-0.7209, -0.7180]; -91.26% , $d = -4.70$). These findings are consistent with the premise of the original simulations, namely that

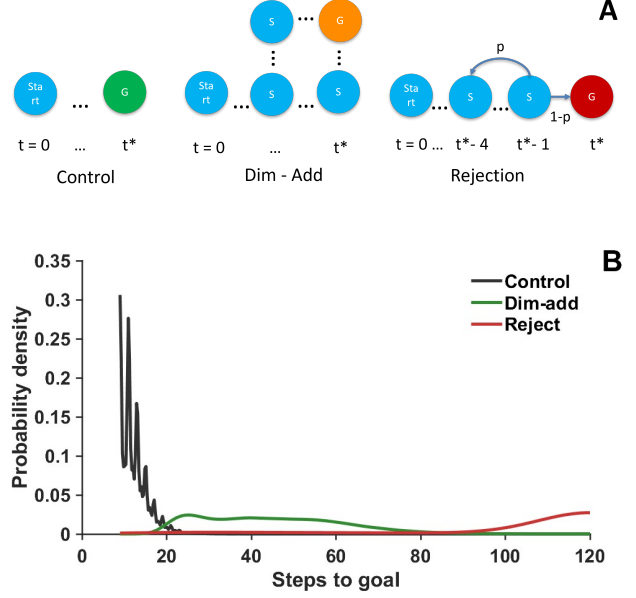


Figure 1: (A): Schematic illustration of the belief space under the three manipulation conditions. In Control, the agent transitions directly from the start state to the goal with a fixed state representation. In Dim-add, an additional dimension of the same length is introduced, expanding the belief space before reaching the same goal and increasing representational complexity without changing the reward and learning structures. In Rejection, belief updates can loop or divert away from the goal state (reflecting the belief that obstacles to a goal are effectively insurmountable), delaying or preventing commitment to the goal despite otherwise identical environmental structure. (B): Population distribution of number of steps to goal states, pooled across all sub-conditions.

adopting belief-space warping reduces the overall level of valenced experience.

Positive and negative hedonic experience dissociate under belief-space warping

Changes in mean hedonic experience were condition-specific; we summarize their relative balance as the *hedonic asymmetry index* (Fig. 2B, D, F). In the Control condition (Fig. 2B), increasing the learning-rate asymmetry parameter A led to a monotonic decrease in the hedonic asymmetry index across all values of the total learning rate S , approaching zero at large A . This pattern indicates that, under ELU alone, increasing asymmetry in learning rates equalizes positive and negative prediction errors rather than producing a strong hedonic imbalance. Similarly, in the Dim-add condition (Fig. 2D), asymmetry indices remained positive but bounded, showing only weaker sensitivity to A . Interestingly, in the Reject condition (Fig. 2F), backward warping produced

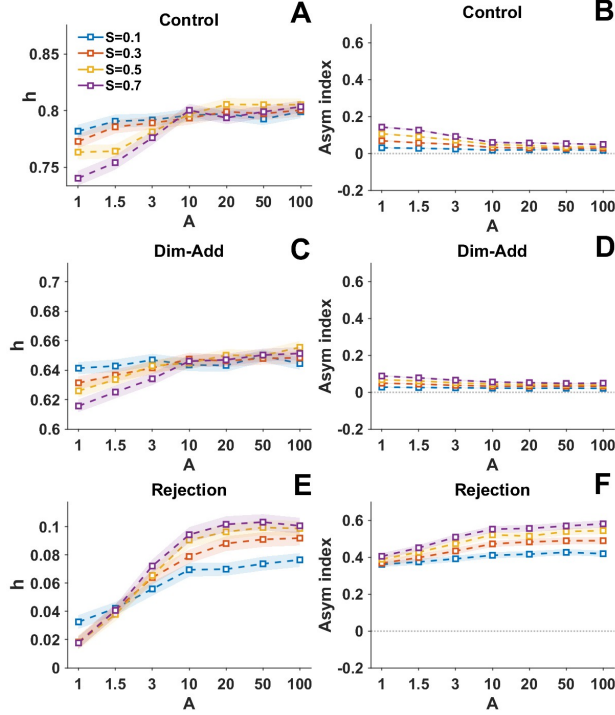


Figure 2: Subject-level distributions of hedonic experience and hedonic asymmetry across conditions, with colored bands indicating standard error. Left column (A,C,E): hedonic experience/ Δ is computed at subject level as a function of negative learning-rate asymmetry (A) and learning-rate mass (S) across all three conditions. Right column (B, D, F): similar construct, with asymmetry indexes of hedonic experience

a large and persistent positive asymmetry index that *increased* with A and S .

Rejection warping asymmetrically reduces negative hedonic experience

To directly quantify the relative reduction of negative versus positive hedonic experience, we computed a ratio $|E[h \mid h < 0]|/E[h \mid h > 0]$ (Fig. 3), which indexes the relative magnitude of negative to positive experience and therefore the level of asymmetry reduction. When pooling across A and S (Fig. 3A), Control showed a stable ratio near unity (mean = 0.9025, CI [0.9014, 0.9037]), indicating comparable scaling of gains and losses. Dim-add produced only a minimal shift relative to Control (mean = 0.9197; $\Delta = +0.0172$, +1.90%, $d = 0.12$), consistent with a mild and largely symmetric rescaling. In contrast, Reject dramatically reduced the ratio (mean = 0.3550; $\Delta = -0.5475$, -60.67%, $d = -1.75$), indicating a pronounced attenuation of negative relative to positive experience.

This pattern was consistent across asymmetry levels (Fig. 3B). In Control, the ratio increased gradually with

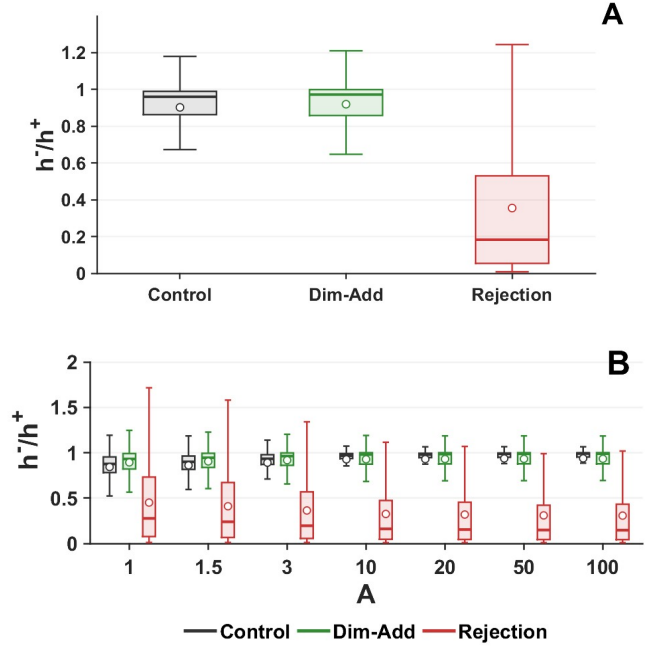


Figure 3: Distributions of negative / positive hedonic experience ratio for all three conditions, computed for individual subjects. Lower ratio indicates more reduction in negative hedonic experience than positive. Boxplots show upper/lower whisker, IQR, and median; circular dots represent mean. (A): Distributions over all simulated subjects, collapsing across A and S . (B): Distributions as a function of ELU levels (A)

A (from 0.842 at $A = 1$ to 0.938 at $A = 100$), reflecting stronger relative weighting of negative outcomes at higher asymmetry. Dim-add closely correlated with Control and converged to near-identical ratios at moderate-to-high A (differences < 0.5%). By contrast, Reject showed a persistent suppression across all A (0.451 at $A = 1$ decreasing to 0.307 at $A = 100$), corresponding to a 46–67% reduction relative to Control (effect sizes $d = -1.14$ to -2.12).

To examine whether these effects are driven by specific stages of learning, we performed a time-window analysis partitioning each episode into early, middle, and late phases (Supplementary Fig. S1). The results show that the strong attenuation under rejection persists across time windows and becomes more pronounced over learning.

The tradeoff between positive and negative hedonic experience

Next, we quantified the coupling between positive and negative hedonic experience by estimating a bootstrapped within-subject tradeoff slope relating $|E[h \mid$

$h < 0$] to $E[h \mid h > 0]$ (Fig. 4). When pooling across learning-rate asymmetry (A), tradeoff slopes differed across conditions as a function of absolute learning rate S (Fig. 4A): Control exhibited the strongest coupling between positive and negative hedonic experience (global mean = 2.546, CI [2.540, 2.552]). Critically, both belief-space warping manipulations substantially reduced this coupling (Dim-add: global mean = 1.188, CI [1.170, 1.207]; Reject: global mean = 1.161, CI [1.147, 1.175]), indicating a global attenuation of hedonic tradeoff under warping conditions. When pooling across S , Control exhibited a strong dependence of the tradeoff on A , with slopes peaking at low-to-moderate asymmetry (e.g., $A=1.5$: mean = 3.249, CI [3.132, 3.367]) and decreasing at higher A (e.g., $A=100$: mean = 1.640, CI [1.489, 1.793]; Fig. 4B). In contrast, belief-space warping flattened both the magnitude and the asymmetry-dependence of this coupling: Dim-add remained near ~ 1 across A (e.g., $A=10$: mean = 0.913, CI [0.829, 1.001]; $A=100$: mean = 0.968, CI [0.876, 1.063]), while Reject was even lower and comparatively stable (e.g., $A=50$: mean = 0.933, CI [0.908, 0.961]; Fig. 4B). A regression model on $\log(A)$ confirmed that Control slopes decreased with increasing A (main effect $\beta_{\log A} = -0.343$, CI [-0.344, -0.342]), whereas both warping manipulations significantly attenuated this negative A -dependence (Dim-add: $\beta_{\text{cond} \times \log A} = 0.293$, CI [0.292, 0.295]; Reject: $\beta_{\text{cond} \times \log A} = 0.365$, CI [0.364, 0.366]). Consistent with the pooled distributions, overall slopes were substantially lower under warping than Control (Control: global mean = 2.406, CI [2.391, 2.420]; Dim-add: global mean = 1.026, CI [1.022, 1.029]; Reject: global mean = 0.931, CI [0.928, 0.933]). The results suggest that belief space warping reduces the tradeoff between positive and negative hedonic experience, with overall learning rate magnitude being an important determinant of the level of reduction. A corresponding time-window based analysis (Supplementary Fig. S2) confirms that the tradeoff patterns persist across learning stages, indicating that the reported effects are not solely driven by full-episode aggregation.

Hazardous environment simulation

To see if the effect of belief space warping is different in reward seeking vs threat avoiding environments, we ran an additional simulation. We simulated learning in a hazardous grid-world of 10 by 10 ("landmine") environment in which agents navigated toward a fixed goal while avoiding randomly distributed high-penalty states that make up 50 percent of the states (each landmine induces a -10 reward, 10 times larger compared to +1/-1 in the world without hazards). All other aspects of the simulation remain the same. Simulations were run on high and low combined learning rates conditions. All metrics were calculated at the subject level and aggregated us-

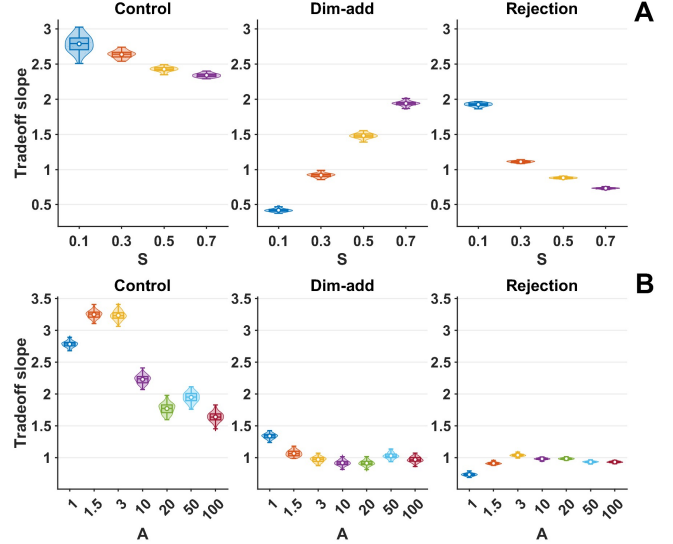


Figure 4: Distributions of the 1000 samples bootstrapped tradeoff slope relating mean negative hedonic experience magnitude to positive hedonic experience conditions. Box-violin plots show the bootstrapped distributions with median, IQR, and min/max. Positive and negative hedonic experience tradeoff is affected by both absolute learning rates magnitude and relative difference. (A): Increasing the total learning-rate scale S systematically modulates the tradeoff within each belief-space manipulation, with higher S amplifying sensitivity to negative outcomes in the Dim-add and Reject conditions while reducing it in Control. (B): When pooled across S , the tradeoff varies non-monotonically with the asymmetry parameter A .

ing bootstrap resampling.

Across both learning-rate scales ($S = 0.1$ and $S = 0.7$, Fig. 5 A and B), the asymmetric hedonic experience increased strongly with the asymmetry of the learning-rate in the Control condition (Fig. 5, fourth column), confirming that ELU induces systematic reduction in hedonic outcomes (h increased with A in Control $S = 0.7$: slope = 0.032, 95% CI [0.0317, 0.0328]), reflecting progressively greater weighting of extreme prediction errors under higher learning rate asymmetry.

Interestingly, the two belief-space warping mechanisms diverged sharply in their behavioral consequences. Dim-Add consistently achieved *higher* cumulative reward than Control across both low and high asymmetry conditions (Fig. 5 first column), albeit with modest effect sizes (e.g., $S = 0.1$: $\Delta = 1.18$, 95% CI [1.13, 1.24], $d = 0.56$; $S = 0.7$, low A : $\Delta = 0.70$, 95% CI [0.60, 0.81], $d = 0.17$), while simultaneously requiring substantially fewer steps to reach the goal (Fig. 5 second column, $S = 0.1$, $\Delta = -143.35$, 95% CI [-149.44, -137.30], $d = -0.60$; $S = 0.7$, $\Delta = -61.62$,

95% CI $[-66.12, -57.04]$, $d = -0.37$). This pattern indicates that dimensional expansion enables agents to exploit a safer and shorter path that improves overall efficiency despite increasing perceived distance to the goal state.

In contrast, Rejection dramatically suppressed hedonic experience (Fig. 5, 3rd and 4th columns), reducing both positive and negative hedonic experience magnitude relative to Control. When pooling across A , Rejection nearly reduced positive hedonic magnitude to 0 in the high learning-rate condition ($S = 0.7$: Control mean = 0.078, CI $[0.077, 0.079]$; Rejection mean = 0.0016, CI $[0.0016, 0.0017]$; $\Delta = -0.077$, CI $[-0.078, -0.076]$, -98%). A comparable suppression was observed for negative magnitude ($S = 0.7$: Control mean = 0.19, CI $[0.19, 0.19]$; Rejection mean = 0.0038, CI $[0.0038, 0.0039]$; $\Delta = -0.18$, CI $[-0.19, -0.18]$, -98%). Although the suppression was less extreme in the lower learning-rate condition, it remained substantial. At $S = 0.1$, positive hedonic magnitude was reduced from Control mean = 0.039 (CI $[0.039, 0.039]$) to Rejection mean = 0.0075 (CI $[0.0056, 0.0097]$), yielding $\Delta = -0.032$ (CI $[-0.034, -0.029]$, -81%). The negative magnitude also decreased from the Control mean = 0.061 (CI $[0.061, 0.062]$) to the Rejection mean = 0.013 (CI $[0.010, 0.018]$), $\Delta = -0.048$ (CI $[-0.051, -0.044]$, -78%).

Together, these results confirm that (1) hedonic asymmetry persists as a direct consequence of ELU, and (2) belief-space warping mechanisms differ qualitatively: dimensional expansion reshapes the state space to enable safer reward acquisition, whereas rejection-based warping primarily suppresses hedonic signals without granting adaptive benefits.

Discussion

We showed, across two different kinds of tasks, that simulated individuals with associative learning mechanisms shaped by high early-life unpredictability (ELU) can benefit by adopting an anhedonic belief about their state space under certain environmental structures. Across two simulation experiments, belief-space warping allowed high-ELU individuals to *improve* their overall hedonic experience by avoiding negative prediction errors, at the cost of reducing some amount of positive prediction errors. These results demonstrate a potential adaptive value to the “warped” mental representations hypothesized to explain anhedonic phenotypes, when adopted in the presence of asymmetric learning rates characteristic of ELU. The framework also makes a new prediction, which is that anhedonia may be particularly adaptive for ELU-experiencing individuals in situations where negative and positive rewards must be adjudicated among – e.g. approach/avoidance or motivational conflict tasks. Previous work has found that anhedonia

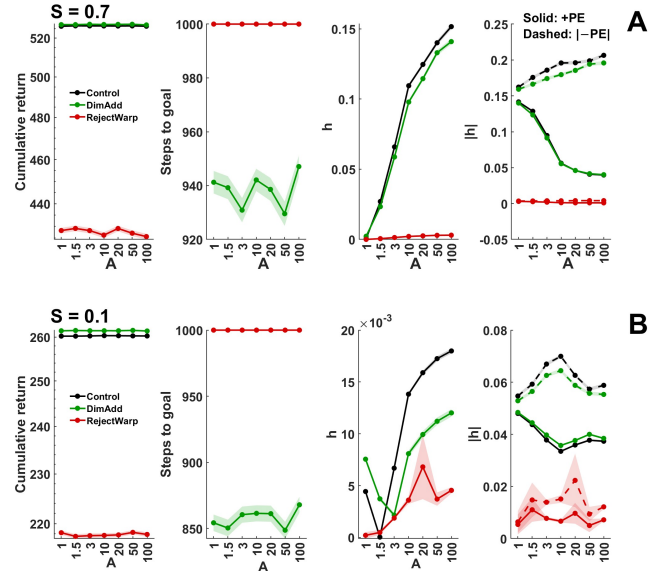


Figure 5: Effects of belief-space warping and learning-rate asymmetry on reward acquisition and hedonic experience. (A): Results for high learning-rate mass ($S = 0.7$). (B): Results for low learning-rate mass ($S = 0.1$). Panels show (left to right): mean cumulative return, steps to reach the goal (lower is better), hedonic experience, and dissociation between positive (solid) and negative (dashed; absolute value) hedonic experience. Shaded regions indicate bootstrapped $\pm$ SEM across simulated subjects.

is widely prevalent in many disorders such as depression and schizophrenia. Future work could adopt this framework to explain how anhedonia may be an adaptive response to symptoms of those disorders, or to comorbid conditions that arise from the same causes.

Importantly, the present model is intended as a minimal computational account rather than a comprehensive model of ELU, anhedonia, or psychiatric disorders. ELU is operationalized as learning-rate asymmetry, and anhedonia as belief-space distortion; both are simplifying abstractions. The gridworld environments are used to isolate computational principles and do not capture the full complexity of real-world cognition, including partial observability, memory processes, or neurobiological mechanisms. Incorporating formal models of each of these factors are important directions for future work.

This approach may also inform treatments. Whereas the original work’s suggestion to focus therapeutic intervention on “unwarping” the structure of belief space might be effective for many individuals, such an approach could be hazardous in the case of high-ELU individuals, as it would reveal the underlying hedonic asymmetry that they learned to avoid. Therefore, our work suggests that different approaches might be needed when these conditions are found together. Importantly,

we find that two distinct forms of belief-space warping can have adaptive value in different conditions: rejection warping is adaptive in pure reward-seeking environments, while dimension-addition can be adaptive in threat-avoidance environments. This insight could guide different forms of behavioral therapy in response to each individual's distinct needs. Future work will examine how the pattern of avoidance and belief-state manipulation predicted here are realized in experimental settings, and whether shifts in the representational geometry of established neural correlates of belief states (Schuck et al., 2016) mediate behavioral changes in line with the model.

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Supplementary Materials

Time-Window analysis of hedonic asymmetry

To examine how hedonic asymmetry evolves over the course of learning, we performed a time-window analysis by taking the shortest common length of episodes and partitioning each episode into three equal segments: early, middle, and late stages. For each segment, we computed the asymmetry index:

$$\frac{E[h \mid h > 0] - |E[h \mid h < 0]|}{E[h \mid h > 0] + |E[h \mid h < 0]|}.$$

This analysis allows us to determine whether the observed effects are driven by specific phases of learning or reflect stable dynamics throughout the episode.

Time-Window analysis of tradeoff between positive and negative hedonic experience

Similarly, to assess whether the tradeoff between positive and negative hedonic experience depends on learning stage, we conducted the same time-window analysis for the positive–negative outcome relationship. For each stage (early, middle, late), we computed:

$$E[h \mid h > 0] \text{ and } |E[h \mid h < 0]|.$$

This analysis examines whether the observed tradeoff between positive and negative experiences is consistent across the course of learning or driven by aggregate effects.

Code Availability

Code for running simulation and data analysis is available at: https://anonymous.4open.science/r/CCN_2026_simulation_and_analysis-8748

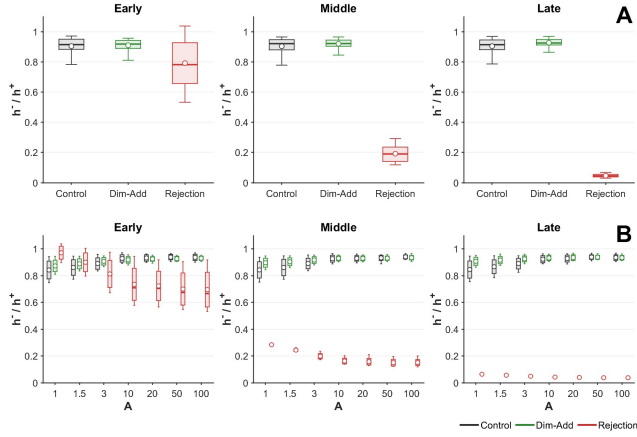


Figure S1: Time-window analysis of asymmetry reduction across conditions. (A) Distributions of the negative-to-positive hedonic experience ratio, $|E[h | h < 0]| / E[h | h > 0]$, pooled across all values of A and S , shown separately for the early, middle, and late stages of learning for each condition (Control, Dim-add, Reject). Lower values indicate greater attenuation of negative relative to positive hedonic experience. (B) The same ratio shown as a function of learning-rate asymmetry A within each learning stage, with conditions overlaid for direct comparison. Across stages, Control and Dim-add remain relatively stable, whereas Reject shows progressively stronger suppression of negative relative to positive hedonic experience over learning.

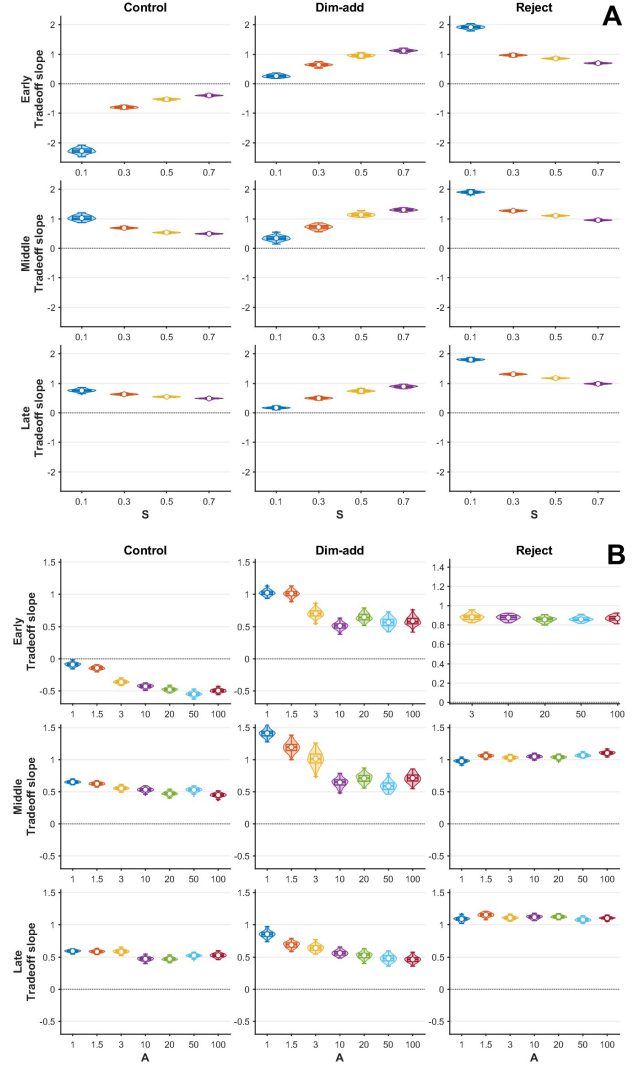


Figure S2: Time-window analysis of the tradeoff between negative and positive hedonic experience. (A) 1000 samples bootstrapped tradeoff slopes relating mean negative hedonic magnitude to positive hedonic magnitude, shown as a function of total learning-rate magnitude S for each condition (Control, Dim-add, Reject) and for the early, middle, and late learning stages respectively. (B) The same tradeoff slopes shown as a function of learning-rate asymmetry A for each condition and learning stage. Together, these analyses show that the main tradeoff patterns reported in the main paper persist across learning stages